GroupPaperB Final Essay

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Executive Summary

A study conducted by the University of California, San Francisco found that 10.5 million United States adults are living with Atrial fibrillation (AFib), up from 5.2 million found in 2010. AFib occurs when the upper chambers of the heart beat irregularly. This can lead to issues such as blood clots, heart failure, and other life-threatening complications. A study conducted showed that 20% of AFib patients lacked confidence in obtaining and interpreting results, leading to a higher chance of complications than others. HeartWatch is an app that allows users to be in control of their health. HeartWatch uses heart rate and electrocardiogram data from external wearables and organizes data in simple graphs. Graph trends and individual health history allow HeartWatch to generate understandable health explanations for users.

HeartWatch improves upon the current market leader, Qaly. Qaly allows users to send an ECG reading to be analyzed by trained professionals. After 30 minutes, users receive feedback about their heart health. However, Qaly offers this service for $30 per month, whereas HeartWatch provides accurate health analysis for free instantly.

The estimated cost for this project is $8,136,485. This includes labor and server costs, Federal Drug Administration (FDA) approval, and data privacy certifications. This app will take 27 months to bring to market. Approximately 9 months will be spent on app development, with the rest spent on clinical trials and receiving FDA approval.

Collecting user data raises privacy concerns. To protect against the risk of data breaches, all HeartWatch data processing will be done using secure servers. Users also can clear any health data off their devices at any time. HeartWatch presents patient health vitals in a simple, comprehensible way, allowing users to be informed of their health.

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1. Introduction/Background

Atrial fibrillation (AFib) is an increasingly common heart condition that primarily affects older adults and individuals with cardiovascular risk factors. AFib is characterized by an irregular and often rapid heart rhythm, which leads to the chaotic beating of the atria (upper heart chambers). This irregular rhythm can result in blood clots forming in the heart, potentially leading to stroke, heart failure, and other serious heart-related complications. This prevalence is expected to grow significantly over the coming decades as the number of people developing AFib increases with age as well as increasing rates of obesity, diabetes, and hypertension. A 2024 University of California, San Francisco (UCSF) study revealed that AFib is three times more common than previously estimated, now affecting 10.5 million Americans or nearly 5% of the population [1]. This finding highlights the growing burden of AFib, fueled by factors such as difficulties tracking symptoms accurately [2]. A 2019 study published by the National Institute of Health (NIH) revealed that inadequate diagnostic and tracking capabilities significantly contribute to the underdiagnosis of AFib. Patients without access to advanced monitoring tools or regular screenings were more than twice as likely to remain undiagnosed compared to those who underwent consistent evaluations (24.6% versus 11.9%) [3]. This diagnostic gap underscores the challenges of accurately identifying AFib, particularly given its intermittent symptoms. Without effective tracking mechanisms, the condition can go unnoticed until complications such as stroke or heart failure arise. Although not always life-threatening on its own, AFib is a significant medical concern that necessitates improved diagnostic strategies to prevent severe outcomes.

2. Proposal Statement

HeartWatch is a Software as a Medical Device (SaMD) designed specifically for users with AFib. It is a software application compatible with any existing smartwatch and mobile device, designed for implementation in wearables like Apple Watch, Fitbit, Pixel Watch, and Garmin products. HeartWatch monitors heart rate, heart pulse, and blood oxygen levels to measure an AFib patient's cardiovascular health, and prompt users when abnormal readings indicate dangerous changes in heart conditions.

This application will allow AFib patients to track their heart status with immediate feedback on their wearable, giving them the insight and information to manage their symptoms. Designed to be cross-functional, this data can be accessed on the application from any phone, tablet, or laptop, organizing and presenting that data for use by customers and their healthcare providers.

3. Work Product

HeartWatch is designed as an integrated application, using data generated from a user's smartwatch wearable that can be accessed from any device. Virtually all modern smartwatches are equipped with a photoplethysmogram (PPG) sensor, a non-invasive technology that uses light to measure blood-volume changes below the skin. These readings turn the pulse, heart rhythm, and blood oxygen level into electrical signals to be analyzed. HeartWatch leverages that existing technology by employing machine learning (ML) to data captured by the wearable to detect irregularities and arrhythmias with 97% accuracy [4]. HeartWatch will use these technologies to keep users safe and informed via vital sign monitoring, real-time alerts and reminders, and detailed data analysis.

Figure 1 - A simulated image of the HeartWatch User Interface for a Pixel Watch (left) and iPhone 16 (right).

Sensor data is the foundation for providing users with active vital sign monitoring for immediate access to their heart health. PPG sensors use green wavelength light emitting diodes (LED) and sensitive photodiodes to measure blood volume beneath the skin. These measurements will be used to compute heart rate, blood pressure, and blood oxygen saturation - essential vital signals used by medical professionals to diagnose and track arrhythmias [5]. The data collected via the PPG is preprocessed to create a representation of the user's heartbeat, the same process a patient undergoing an electrocardiogram (ECG) would receive at a doctor's office, but on their smart device. *Figure 2* illustrates the refinement process from raw user data to a heartbeat or heart pulse that can be analyzed further to detect arrhythmias.

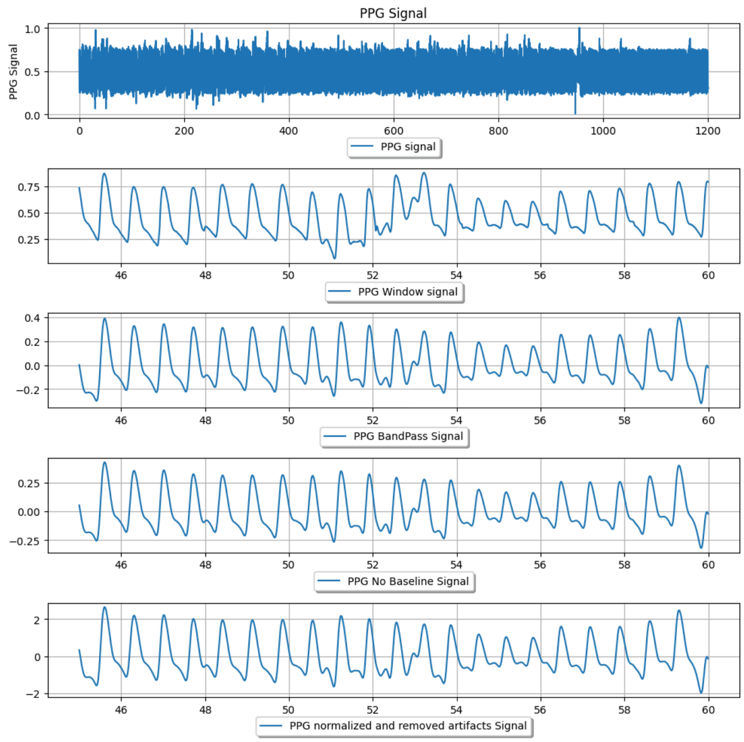

Figure 2 - Raw PPG data refined and normalized to be interpreted by the ML stack [4].

In addition to live vital monitoring, users will need to know if abnormal vitals approach a dangerous threshold or indicate an ongoing Afib episode. The ECG signal will be analyzed to observe if users' heart data is in extended periods of high or low blood pressure, elevated or depressed heart rate, and arrhythmic pulse period. If these events are detected, the application will inform them that they are experiencing a cardiac event. HeartWatch will indicate the severity and duration of these events, record their characteristics, and indicate if a medical professional should be alerted.

By comparing user data to model sets of healthy sinus rhythm (heartbeat), the application will employ a stack of machine learning (ML) algorithms to evaluate those signals. Stacking is a method in ML that combines the predictive outputs from multiple base models to determine a final prediction with greater overall accuracy than the individual models. Three stacked data collection models will be used to classify PPG data. Support Vector Machine (SVM), Random Forest (RF), and Multi-layer Perceptrons (MLP) are methods that are over 97% effective in detecting arrhythmias and abnormal changes in blood pressure when compared to existing Federal Drug Administration (FDA) approved devices [4]. SVM and RF are classification algorithms used to determine the probable output of a system. In this case, they are the primary drivers in determining whether a pulse parallels healthy or arrhythmic patterns. MLP is a neural network that supplements the classification of PPG signals. These methods stacked together provide the user with accurate information to detect cardiac events, but they also allow HeartWatch to store and organize this data to be used by a patient's primary care physician or cardiologist.

HeartWatch's cross-functionality means that while users are monitored in real-time, their heart data is being collated for observation in the application from any of their devices. As it is a diagnostic tool that users rely on for accurate medical measurements, it will be classified as a Class II FDA-compliant device. Similarly, as HeartWatch tracks, processes, and stores user data, it will observe all HIPAA compliance requirements to protect the customer's information. This data provides longer-term trend information to users, while simultaneously allowing doctors to see their patient's heart health between visits. Data is organized by date and time and can be downloaded by a user to be shared with their cardiologist. Because Afib is highly manageable with consistent supervision, HeartWatch fills in the gaps for users and doctors with crucial Afib history that would otherwise be lost between visits.

HeartWatch provides real-time insights for individuals managing AFib and supports healthcare institutions like professionals in hospitals and caregivers at home, by offering continuous monitoring and early detection of abnormal events. The integrated system brings possible medical concerns to the attention of the user and their healthcare professional for timely intervention. HeartWatch emphasizes data communication and professional partnerships, empowering both patients and healthcare providers to make informed decisions for improved health outcomes.

4. Target Customer

UCSF shattered preconceived notions about the extent of Afib diagnoses in the United States. The study, which appears in the Journal of the American College of Cardiology, estimates that there are three times as many people living with Afib than previously thought [1]. The study indicated that throughout the research, “AF patients became relatively younger, were less likely to be female or White, and were more likely to have hypertension and diabetes”, reflecting a wider population is at risk than previously assumed. HeartWatch intends to service the 10.55 million Americans currently estimated to be living with Afib. As more people are diagnosed with Afib younger, the application will serve a population that is more technologically proficient, while giving older patients a simple solution to manage their symptoms.

To contextualize broadly, as of 2021 the global market for medical health apps was estimated to be valued at US$36.1 billion [6]. Growth in this sector is projected at a 14.3% annualized rate, meaning a total financial opportunity of US$110.7 billion over the decade spanning 2022-2032 [6]. HeartWatch can be considered a diagnostic app, whose sector is estimated to grow at an annualized rate of 15.9%, thus outperforming the broad medical app market between 2022 and 2032. Surveys of smartwatch users indicate that 82% use fitness and health tracking features, and 79% use notifications and alerts [7]. Statista and the Pew Research Center report that 15% of smartwatch users are adults aged 55 and up, and are specifically motivated to use technology to track and monitor health. HeartWatch aims to ride this broader market momentum while targeting Afib patients specifically. This indicates an opportunity in a rapidly growing sector, and positions HeartWatch to be a leader against its competition.

5. Competition

The main competitors to HeartWatch are the Qaly ECG Reader and Apple Health apps. Similar to HeartWatch, Qaly integrates with existing hardware such as Apple Watch and Fitbit to read patient ECG. Users then have the option to send the ECG readings to certified experts and shortly receive a response [8]. However, there are a few drawbacks to the Qaly app. First off, Qaly's method of vital analysis takes around 30 minutes to provide users with information about their health, whereas HeartWatch provides instant vital analysis at the same level of accuracy. Figure 3 shows Qaly's app interface and an example of an expert's ECG interpretation. Additionally, Qaly requires users to subscribe for $30 per month to get access to certified ECG reviews whereas HeartWatch offers unlimited vitals tracking at no cost [8]. AFib patients already struggle with access to medical information. Qaly's subscription model and non-immediate results still make it difficult for AFib patients to obtain information about their health. HeartWatch's free, immediate results are exactly what users need to have a straightforward understanding of their health.

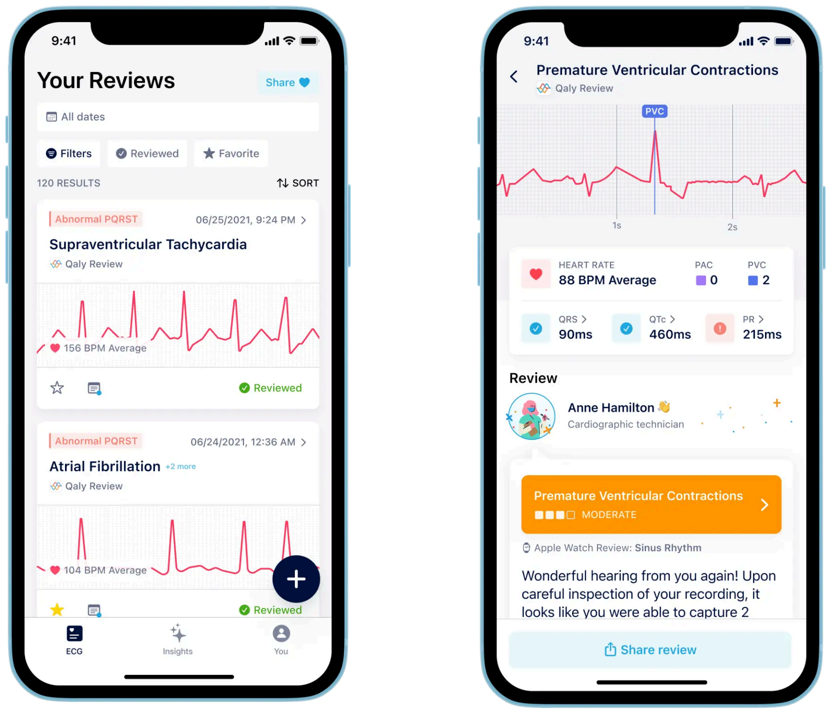

Figure 3 - Qaly user interface showing a history of ECG readings (left) and expert analysis of ECG reading (right) [8].

Apple Health is also a direct competitor to HeartWatch. Apple Health works only with the iPhone and Apple Watch. Vitals such as heart rate and ECG are taken on the Apple Watch, and all vitals are available on the app, along with short explanations of the results, as shown in Figure 4 [9]. This figure also shows that the Apple Health app automatically alerts users when irregular rhythms are detected. While the main features of Apple Health and HeartWatch are similar, Apple Health is severely limited by its lack of support for non-Apple devices. 53.1% of individuals in the United States currently own iPhones [10]. For non-Apple users, the main alternative is Qaly, which is available on both Android and Apple devices. However, the downsides of Qaly have been mentioned above. HeartWatch is available to download on any device such as phones, tablets, and laptops to provide users with the ultimate convenience for their heart health.

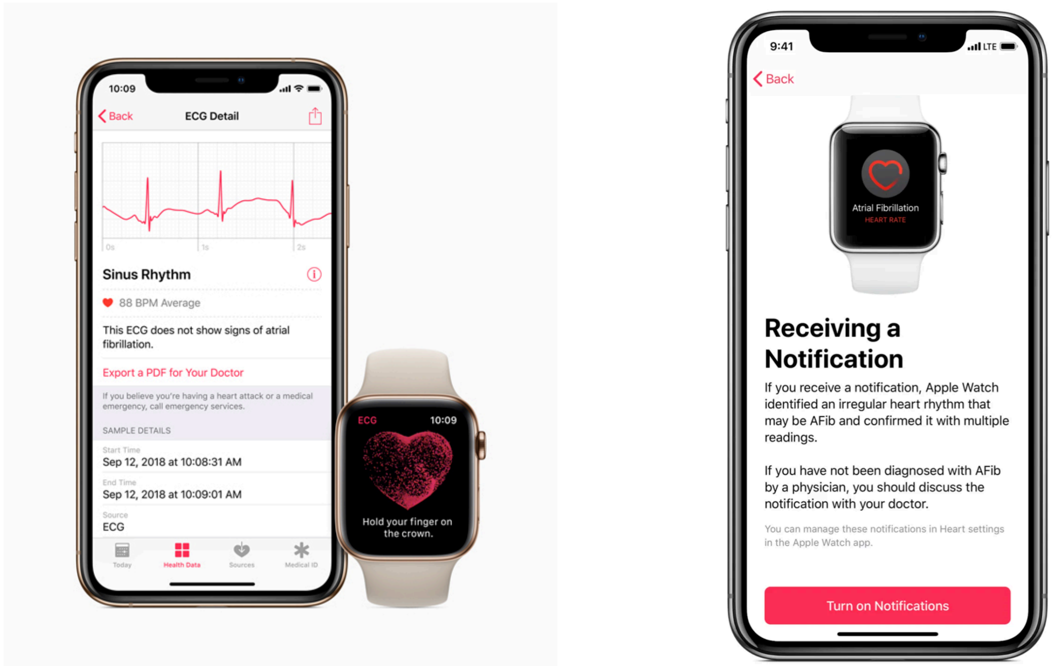

Figure 4: User interface of Apple Health showing ECG reading (left) and irregular heart rate notification setup (right) [9].

6. Timeline and Costs

The total time expected to bring this app to market is 27 months. There are two main parts of the development process. The first part will focus on app development, with the second part focused on testing app effectiveness to obtain FDA approval. As seen in Table 1, app design and feature development will occur simultaneously, taking a total of six months. After the groundwork of features has been developed, testing and iterative app updates will follow for three months [11]. The testing and updates will be focused on debugging software to avoid a poor user experience.

Clinical trials will follow to confirm the efficacy of the app’s vital interpretation and feedback. Adhering to FDA guidelines, clinical trials will occur in three phases and take a total of one year to complete. Phase one of the trials will consist of a group of 50 volunteers and will take three months. Phase two will then expand the group to several hundred people with a similar timeline to phase one. Finally, phase three will further expand the trial group to 1,000 individuals and will last six months [12]. The FDA clearance application will then be submitted, with an estimated six months until approval [13]. The app is then expected to be released during the 27th month.

Table 1: Timeline	Month 1-3	Month 4-6	Months 7-9	Month 10-12	Month 13-15	Month 16-18	Month 19-21	Month 22-24	Month 25-27
App Design									
Backend/Feature Development									
Stability testing									
Iterative Updates									
Trial Phase 1									
Trial Phase 2									
Trial Phase 3									
FDA Approval									
Finalize and Release									

Table 1: Timeline of complete app development and release.

The total cost to bring this app to market is $8,136,485. The main costs associated with bringing this project to market include trial testing, regulation approvals, labor costs, and server fees. Four total app developers will be hired with one focused on design, another on feature development, and two on quality assurance. The average yearly salary for one app developer is $132,000 and $98,500 for a project manager [14,15]. Table 2 breaks down these salaries on a dollar-per-month basis and multiplies this rate by the contract length for each employee based on the Table 1 timeline. The duration reflected in Table 2 reflects the need to bring back the feature developer and quality assurance engineers to update the app based on feedback from clinical trials. All these totals are then added up and equate to a total of $617,616 for labor costs.

Labor Type	Monthly Rate ($/month)	Duration (months)	Total
Design Interface Developer	$11,000	6	$66,000
App Feature Developer	$11,000	12	$132,000
Quality Assurance 1	$11,000	9	$99,000
Quality Assurance 2	$11,000	9	$99,000
Project Manager	$8,208	27	$221,616
Total			$617,616

Table 2: Total labor costs with breakdown.

There are costs other than labor that also need to be considered. It will take 7.5 million dollars to complete three phases of clinical trials [16]. These funds will be used to compensate participants, purchase wearables for trials, and labor costs for employees in charge of trials. The FDA clearance application is $6,084, with HIPAA certification amounting to $1,040 [17,l8]. HIPAA certification is necessary to maintain patient privacy since HeartWatch shares patient health with doctors. For the app to run, it will also need to be hosted by a server such as Amazon Web Services Relation Database (RDS) Amazon RDS allows for quick data transfer speeds under high loads, and sorts all vital data automatically into a secure database. The cost associated with Amazon's RDS for the duration of development is $11,745, with a detailed breakdown available in Table 3 [19].

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Cost Source	Rate	Total
Clinical Trials	N/A	$7,500,000
FDA 510k Application	N/A	$6,084
HIPAA	$1,040/yr	$1,040
Amazon Web Services RDS	$435/month	$11,745
		$7,518,869

Table 3: Breakdown of the cost of services.

7. User Risk Management

The most important concern in HeartWatch is the risk of undetected AFib events, especially for users unfamiliar with technology, those who may misuse the app, or misread critical notifications. To address this, HeartWatch ensures that key notifications about health abnormalities reach both the user and their designated connected devices, such as those of family members and caregivers, particularly in emergencies. This ensures that even if the user does not respond to alerts, caregivers can assist in initiating appropriate health measures.

Overdependence could be another issue. Although current machine learning algorithms achieve high detection rates, it is still crucial for users to stay aware of their symptoms. HeartWatch will recommend users consult their doctors based on their health data rather than relying solely on automated alerts. HeartWatch will notify users to follow healthcare instructions and inform doctors or caregivers if regular check-up visits are missed. To provide users with actionable and trustworthy notifications, HeartWatch ensures that critical health data is accurate and readily shareable with medical professionals, enhancing trust and usability for users and caregivers.

The last concern is data security, as many people today still underestimate the importance of privacy [20]. To protect sensitive health information HeartWatch will incorporate multiple layers of security, including strong password protection, multi-factor authentication, and encryption of both stored and transmitted data. HeartWatch will comply with the HIPAA Privacy Rule to meet industry standards for health data protection. A secure data storage server, such as Amazon RDS, will be used to ensure robust data protection. Additionally, users will have control over their data, as the model will run locally. Users can choose whether to store information in the cloud and clear their data as needed. HeartWatch will also provide clear guidelines and educate users through tutorials, practical examples, and periodic updates. Recognizing that data security is a shared responsibility, the goal is to build user trust by fostering awareness of best practices and ensuring personal health information is managed securely, adhering to industry-leading standards.

8. Conclusion

HeartWatch aims to address the challenges for those suffering from an Afib diagnosis. By offering continuous monitoring, real-time notifications, and actionable health insights, HeartWatch empowers users and healthcare professionals to make informed decisions and take timely actions. HeartWatch integrates seamlessly with existing wearable devices, using advanced machine learning to detect irregularities in cardiovascular health. The collaboration with healthcare institutions and compliance with FDA standards ensures that HeartWatch adheres to the highest levels of medical accuracy, making HeartWatch a reliable tool for both patients and medical practitioners.

With its focus on data security and privacy, HeartWatch offers a trustworthy platform that safeguards sensitive health information while fostering user awareness of the importance of data handling. The directive for HeartWatch is to improve health outcomes, reduce healthcare burdens, and provide a user-friendly means for managing AFib effectively.

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